DOC-23 | CONTROLLED SOURCE SYNCHRONIZATION

Cross-reference-only revision notice

v1.0-RC3 | 11 September 2026 | SOURCE-OF-TRUTH SYNCHRONIZATION ONLY

Control	Selected value
Revision	v1.0-RC2 -> **v1.0-RC3**
Semantic change	**NONE.** This revision changes source version/count/path references only.
Wave-2 semantic authorities	DOC-07 RC2 | DOC-08 RC2 | DOC-15 RC8 | DOC-16 RC5
Canonical HTTP companion	contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml
OpenAPI SHA-256	bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00
RC8 estate	113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL

DOCUMENT-SPECIFIC SYNCHRONIZATION

• Repository/technology architecture is unchanged. The executable HTTP source is now DOC-15 v1.0-RC8 at contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml.

• Runtime/client parity gates use the RC8 estate: 113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs; 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL.

• Generated TypeScript/Python transport artifacts must continue to derive from the canonical YAML. FastAPI-generated OpenAPI remains non-canonical. No application, package, dependency direction or technology choice changes.

SUPERSESSION AND HISTORICAL CLASSIFICATION

The inherited body that follows is retained unchanged in substantive page content. Any inherited non-owning reference to DOC-15 RC7/RC6, the old 112/85/217/1,297 estate, 11 AUTH operations, DOC-16 RC4, or the RC7 YAML filename is classified **HISTORICAL / SUPERSEDED REFERENCE** and is not normative in this revision. All other inherited owned semantics remain in force.

Occurrence ledger: RC7/old HTTP: inherited printed pages 1, 3, 4, 5, 8, 17, 19, 20, 21, 22, 23, 24, 25, 27; RC4: inherited printed pages 1, 20.

IMPLEMENTATION IMPACT: NONE. The accepted NAKSHATRA_MASTER_WAVE2_HANDOFF application/database/API behavior is unchanged by this document synchronization.

BLOG-DOC-23

Technical Architecture, Stack, Repository & Dependency Contract

Revised Technology Baseline

Engineering codebase authority for a mixed-language Hybrid Monorepo: 14 isolated Next.js public applications, one React/Vite Central Admin, one Python/FastAPI API, one Python Worker, shared contracts, database assets, infrastructure, tests, tooling and documentation.

14	1	1	1	2
public site apps	Central Admin	API app	Worker app	language ecosystems

CORE RULE	One repository does not mean one frontend or one runtime. Every public site remains independently editable, buildable, testable and deployable. Product behavior and upstream semantic contracts do not change because the implementation language changes.

Document control	Value
Status	REVIEW CANDIDATE - revised implementation baseline; freeze only after cross-document audit and project approval
Version	1.0-RC2
Date	10 September 2026
Supersedes	BLOG-DOC-23 v1.0-RC1 implementation stack only; all preserved non-technology architecture rules remain in force
Upstream baseline	DOC-01..14; DOC-15 RC7 + canonical OpenAPI; DOC-16 RC4; DOC-17 RC2; DOC-18 RC2; DOC-19 RC2; DOC-20 RC1; DOC-21 RC1; DOC-22 RC1
Primary authority	Implementation technology, mixed-language repository structure, code/application/package boundaries, dependency/import rules, naming, deterministic locking, affected-build orchestration and scaffold strategy
Explicit non-authority	No change to product scope, permissions/capabilities, Page IDs, API operations/schemas/statuses, DB tables/columns/RLS, routing, SEO/search algorithms, media variants, worker job semantics, security policy, SLO/RPO/RTO or QA behavior
Implementation evidence	NOT_VERIFIED / EVIDENCE_REQUIRED until a candidate repository, build and QA evidence set exists

DEVELOPER LOCK	Do not reintroduce Astro, Fastify, Kysely, node-postgres or BullMQ into the canonical backend/public baseline; do not use Next.js/FastAPI/Celery defaults to redefine frozen behavior; do not make ORM metadata, framework-generated OpenAPI, broker internals or shared visual packages a second source of truth.

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B	Package/import dependency matrix
C	Per-site Next.js page-local template
D	Python layer map
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1. Purpose, Authority and Revision Rules

BLOG-DOC-23 RC2 replaces only the implementation-technology choices of RC1. The Hybrid Monorepo, fourteen independent public application model, Central Admin topology, shared core, SiteContext isolation, dependency inversion, site-local visual ownership and no-cross-site-import rules remain architectural invariants.

- PDF-23 owns HOW code is organized and which implementation stack is used; it does not move semantic authority out of PDFs 01-22.
- Framework convenience is subordinate to DOC-15 OpenAPI, DOC-08 PostgreSQL/RLS, DOC-18 job/cache mechanics, DOC-17 media behavior and DOC-19 security rules.
- A technology that cannot satisfy an upstream rule is rejected or explicitly classified as a blocker; upstream semantics are not silently weakened.
- Nakshatra technology experience is supporting engineering context only. No Nakshatra product feature, database object, permission, workflow or business rule is imported into this Blogging Platform.

REVISION RULE	RC1 implementation references are superseded where they conflict with this RC2. Preserved architectural decisions from RC1 continue unless RC2 explicitly replaces the technology mechanism.

2. Frozen Invariants and Superseded Implementation Decisions

ID	Invariant preserved in RC2
ARC23-INV-01	Every ordinary site-owned backend operation receives exactly one trusted SiteContext before site-owned data access.
ARC23-INV-02	Public Host, Admin site selection and worker payload identity are inputs to trusted context resolution, never authorization by themselves.
ARC23-INV-03	No frontend connects directly to PostgreSQL, Redis, queue broker, OpenSearch, private object origin or security-state store.
ARC23-INV-04	No site app imports another site app directly, indirectly, through aliases, re-exports, symlinks or generated code.
ARC23-INV-05	No shared frontend package owns public visual components, theme, Header/Footer, navigation UI, assets or mutable current-site state.
ARC23-INV-06	DOC-15 RC7 OpenAPI is the executable HTTP authority; generated frontend types and FastAPI implementation cannot redefine it.
ARC23-INV-07	DOC-08 raw SQL migrations remain authoritative; SQLAlchemy metadata/schema generation is not migration authority.
ARC23-INV-08	API and Worker are separate deployables that share Python domain/application packages but never import each other's entrypoints.
ARC23-INV-09	Shared backend code contains no SITE-01..SITE-14 behavior switch except non-authoritative fixtures/manifests.
ARC23-INV-10	Future sites add one site-local public app and registry/config work; API, Worker, DB, Redis and search are not cloned.

RC1 implementation choice	RC2 disposition
Astro SSR public sites	SUPERSEDED by Next.js + React + TypeScript.
Fastify + Ajv API	SUPERSEDED by Python 3.13 + FastAPI + Pydantic, with DOC-15 RC7 unchanged.
Kysely + node-postgres	SUPERSEDED by SQLAlchemy 2.0 + Psycopg 3; raw SQL migrations remain.
ioredis in backend	SUPERSEDED by redis-py in Python backend/worker; Redis server/key semantics unchanged.
BullMQ worker/broker adapter	SUPERSEDED by Python worker using Celery execution plus a durable delayed-work adapter on dedicated Redis broker plane.
Node crypto/otplib/Nodemailer/Pino	SUPERSEDED by Python-compatible adapters without changing DOC-19/20 semantics.
One repository lockfile	REFINED to two authoritative ecosystem lockfiles: pnpm-lock.yaml and uv.lock.

3. Revised Technology Stack Baseline

Concern	Frozen line	Boundary / required use
Public runtime/build	Node.js 24.x	For Next.js site servers/build only; not API/Worker runtime.
Public framework	Next.js 16.3.x + React 19.x + TypeScript 5.x	14 independent SSR-capable applications; exact patch locked.
Admin	React 19.x + TypeScript 5.x + Vite 7.x + React Router 7.x + TanStack Query 5.x	One Central Admin; 19 modules/36 operational pages unchanged.
JS workspace	pnpm 10.x + Turborepo 2.x	Frontend/tooling workspace and top-level affected task graph.
OpenAPI TS	openapi-typescript 7.x	Generated wire types from DOC-15 RC7 YAML.
Backend runtime	Python 3.13.x	API/Worker shared language baseline.
API	FastAPI 0.141.x	HTTP adapter only; canonical OpenAPI remains DOC-15 RC7 file.
Validation	Pydantic 2.13.x	Transport/config/internal validation; cannot broaden wire schemas.
DB access	SQLAlchemy 2.0.x stable + Psycopg 3.3.x	Repository/access layer only; DOC-08 SQL/RLS semantics remain canonical.
Python deps	uv + PEP 621 pyproject.toml	Exact resolved Python graph locked in uv.lock.
Control/cache Redis	Redis 8.x + redis-py 8.1.x	DOC-18 keys/TTL/lease/dedupe/cache semantics.
Queue execution	Celery 5.6.x	Python task execution mapped to DOC-18 logical queues/jobs; no Celery semantic authority.
Queue broker	Dedicated Redis 8.x broker plane	Separate credentials/persistence/memory policy from control/cache Redis.
Search	OpenSearch 3.x + opensearch-py 3.x	Derived only; exact DOC-16 ranking and site scope.
Object storage	S3-compatible + boto3/botocore	Private original + fixed derivative delivery per DOC-17.
Image processing	libvips via pyvips (baseline)	Fixed DOC-17 transforms/sanitization only; no dynamic transform API.
Security crypto	argon2-cffi + PyOTP + Python secrets/hashlib/hmac/cryptography	Exact DOC-19 policy and parameters.
Security state	DynamoDB-compatible conditional/TTL adapter via boto3	Separate from DOC-18 Redis.
Secrets	Vault-compatible adapter (hvac/HTTPS)	External secret/envelope boundary.
Mail	SMTP adapter (aiosmtplib or equivalent)	Invitation/reset delivery only.
Observability	OpenTelemetry Python + structured JSON logging + prometheus-client	DOC-20 semantic dimensions/retention remain authority.
Python tests	pytest + pytest-asyncio + testcontainers-python	Backend/unit/integration.
Browser/A11y	Playwright + axe-core	Public/Admin E2E and accessibility.
Load/DR	k6 + runbooks/scripts	DOC-20/21 performance and DR evidence.
Static quality	ESLint/Prettier/dependency-cruiser + Ruff/mypy/Import Linter	Enforce TS and Python boundaries.
Local infra	Docker Compose (compose.yaml)	Disposable dependencies only; production wiring remains DOC-24.

VERSION POLICY	Major/minor technology lines are frozen here. Exact patch versions and transitive versions are locked by pnpm-lock.yaml and uv.lock. No arbitrary patch number becomes prose authority.

4. Technology Rationale and Compatibility Gate

Technology	Why selected	Critical compatibility rule
Next.js + React	React-first UI richness with server rendering, per-site deployments and mature routing/build tooling.	Use SSR route control without creating a second routing/SEO authority.
Python + FastAPI	Strong typed request handling, async I/O and a clear transport/application split.	FastAPI-generated OpenAPI is diagnostic only; it must compare equal to DOC-15 where relevant and never replace it.
SQLAlchemy + Psycopg	Modern Python DB access, explicit transactions and	No schema push/create_all; all tenant RLS

Technology	Why selected	Critical compatibility rule
	asyncio support.	transactions use trusted SET LOCAL.
Celery + Redis broker	Mature Python worker execution and routing, compatible with Python 3.13.	Far-future scheduling, deterministic retry/dedupe and DLQ semantics require the explicit adapter controls in Section 17; Celery defaults are insufficient.
OpenSearch Python client	Language-native adapter for the already-selected derived search engine.	Provider ranking never overrides DOC-16 deterministic ordering.
pyvips/libvips	Language-native high-performance image adapter.	Outputs must be byte/semantic-equivalent to DOC-17 variant contract; no new variant.

Compatibility verification at this revision date confirms: Next.js 16.3 is the active line and requires Node.js 20.9 or later; Python 3.13 is supported by the chosen SQLAlchemy 2.0/Psycopg 3/Celery lines; SQLAlchemy 2.1 is still a release candidate and is therefore not the baseline. Exact production-resolved versions remain lockfile authority.

5. Hybrid Monorepo and Mixed-Language Workspace

```
/
  package.json                # JS/TS orchestration root
  pnpm-workspace.yaml
  pnpm-lock.yaml              # authoritative JS/TS lock
  turbo.json
  pyproject.toml              # Python workspace/project policy
  uv.lock                     # authoritative Python lock
  apps/                       # 14 sites + admin + api + worker
  packages/ts/                # approved TS contracts/clients/pure utilities
  packages/python/            # domain/application/ports/adapters
  contracts/http/             # DOC-15 RC7 canonical OpenAPI
  database/                   # DOC-08 raw SQL migrations/seeds/RLS verification
  infrastructure/ tests/ tooling/ docs/
```

- `pnpm-lock.yaml` is authoritative for JavaScript/TypeScript dependencies; `uv.lock` is authoritative for Python dependencies. Neither replaces the other.
- Package-local npm/yarn/pnpm lockfiles and package-local Python lockfiles are prohibited. `requirements.txt` may be generated for deployment only and must be reproducible from `uv.lock`; it is not an independent authority.
- Turborepo remains the top-level task orchestrator. Python API/Worker task nodes are exposed through tiny private package.json bridge manifests that delegate to `uv run`; Node is an orchestration dependency there, not a backend runtime.
- Python import boundaries are enforced independently with Import Linter/mypy/Ruff. Turborepo dependency edges do not substitute for Python import-graph enforcement.
- Workspace globs discover `apps/sites/*`; no CI/build script encodes a lifetime ceiling of fourteen sites.

6. Deployable Application Registry

App class	Root	Initial count	Runtime	Locked rule
APP-PUBLIC-*	apps/sites/site-*/	14 + future	Node 24 / Next.js 16.3	One site presentation per app; independently built/deployed.
APP-ADMIN	apps/admin/	1	Static Vite output	One Central Admin; no public site visual imports.
APP-API	apps/api/	1 logical app	Python 3.13 / FastAPI	Stateless shared HTTP app; horizontally replicated.
APP-WORKER	apps/worker/	1 code app	Python 3.13 / Celery worker + scheduler adapters	Six queue profiles may scale independently without new app/job semantics.

NO SERVICE DRIFT	Python packages for posts/media/search/auth remain modules of the shared backend core, not microservices. API and Worker are the only backend application deployable classes defined here.

7. Canonical 14-Site Application Registry

App	Site	Current domain	Folder	Package
APP-01	SITE-01	WhatsYourRashi.in	site-01-whats-your-rashi	@bp/site-01-whats-your-rashi
APP-02	SITE-02	ApkiRashi.com	site-02-apki-rashi	@bp/site-02-apki-rashi
APP-03	SITE-03	DailyRashi360.com	site-03-daily-rashi-360	@bp/site-03-daily-rashi-360

App	Site	Current domain	Folder	Package
APP-04	SITE-04	ZodiacDaily.in	site-04-zodiac-daily	@bp/site-04-zodiac-daily
APP-05	SITE-05	DailyHoroscope360.com	site-05-daily-horoscope-360	@bp/site-05-daily-horoscope-360
APP-06	SITE-06	MyHoroscope360.com	site-06-my-horoscope-360	@bp/site-06-my-horoscope-360
APP-07	SITE-07	AstroToday360.com	site-07-astro-today-360	@bp/site-07-astro-today-360
APP-08	SITE-08	ApkiKundli.com	site-08-apki-kundli	@bp/site-08-apki-kundli
APP-09	SITE-09	MatchYourKundli.com	site-09-match-your-kundli	@bp/site-09-match-your-kundli
APP-10	SITE-10	ApkaNakshatra.com	site-10-apka-nakshatra	@bp/site-10-apka-nakshatra
APP-11	SITE-11	JaniyePanchang.com	site-11-janiye-panchang	@bp/site-11-janiye-panchang
APP-12	SITE-12	RashiRemedies.com	site-12-rashi-remedies	@bp/site-12-rashi-remedies
APP-13	SITE-13	ApkaPandit.in	site-13-apka-pandit	@bp/site-13-apka-pandit
APP-14	SITE-14	VastuShastra360.in	site-14-vastu-shastra-360	@bp/site-14-vastu-shastra-360

Folder/package identity remains anchored to stable SITE/APP identity plus a readable slug. Runtime tenant authority is always trusted Host -> SiteContext, never the package name or BP_SITE_APP_KEY.

8. Next.js Public-Site Folder Contract

```
apps/sites/site-01-whats-your-rashi/
  package.json
  next.config.ts
  tsconfig.json
  public/                        # SITE-01 static files only
  src/
    pages/                       # thin Next.js Pages Router SSR adapters
      index.tsx
      article/[post_slug].tsx
      category/[category_slug].tsx
      tag/[tag_slug].tsx
      author/[author_slug].tsx
      search.tsx
      about.tsx
      contact.tsx
      privacy-policy.tsx
      terms-of-use.tsx
      disclaimer.tsx
      _preview/post/[post_id].tsx
      404.tsx
      _app.tsx
      _document.tsx              # only if site needs controlled document shell
    site/                        # ALL presentation ownership for this site
      site-app.manifest.ts
      shell/header/ footer/ navigation/
      pages/home/ article/ category/ tag/ author/ search/
      pages/static-about/ static-contact/ static-privacy/
      pages/static-terms/ static-disclaimer/ preview/ not-found/ gone/
      components/ styles/ assets/ lib/ tests/
```

- The same physical ownership pattern applies to APP-01..APP-14. Source is never imported across site packages.
- `src/pages/` owns route/SSR adaptation only. Page-specific composition, components, styles and tests live under `src/site/pages/<page>/`.
- Header, Footer and navigation UI remain under that site's `shell/`. There is no shared `@bp/ui`, `@bp/theme`, `@bp/header`, `@bp/footer`, `@bp/cards` or site-shell runtime package.
- `public/` and `src/site/assets/` cannot symlink/alias another site's directory. Fonts and other visual source assets are site-local.
- Next.js `app/` Router is not the canonical public routing layer in this baseline. Pages Router SSR is intentionally selected because exact upstream HTTP status outcomes must be set deterministically per request.

9. Public SSR, Routing and Status-Code Contract

Each DOC-09 route family is implemented as a thin Next.js Pages Router page. `getServerSideProps` (or the exact equivalent supported by the pinned Next.js line) calls the shared public API using the effective request Host and returns presentation props or an exact route outcome. The frontend never recomputes tenant or canonical authority.

Upstream result	Next.js adapter behavior	Prohibition
200 PageData	SSR local site page module from DOC-13 PageData.	No direct DB/search/storage fetch.
301/302/308 redirect	Set exact redirect status and Location from API/route authority.	Do not use framework-preferred status if it differs.
404	Return framework notFound or exact 404 rendering mapped to frozen Not Found semantics.	No cross-site fallback.
410	Set `res.statusCode = 410` and render local Gone module.	Do not convert to 404 or invent /410 route.
Preview	Call API15-PUB-002 with host/site/post-bound Preview session; no-store/noindex.	Never put grant/session in PageData.

```
request Host
  -> site Next.js SSR adapter
  -> @bp/api-client-public
  -> API15-PUB-001 or API15-PUB-002
  -> FastAPI resolves trusted Host -> SiteContext -> PageData/status
  -> adapter preserves status/Location/cache/no-store
  -> site-local React presentation renders DOC-12 semantics
```

UNKNOWN HOST	Receiving traffic at a particular Next.js deployment never authorizes that site. Unknown Host must preserve DOC-09 neutral handling; APP/SITE build metadata is diagnostic only.

10. Next.js Rendering, Data, Cache and Client Boundaries

Boundary	Locked behavior
Server rendering	Canonical public pages are server-rendered per request when DOC-11/13 current state/status is required. Static optimization may not bypass current PageData/route authority.
Data fetching	Server adapter uses the generated public client; all content/SEO/navigation/media comes through DOC-13/DOC-15 projections.
Next data cache	Authoritative PageData fetches use no framework semantic cache. DOC-18 Redis and DOC-17/CDN remain the cache authorities. A framework build/cache may cache immutable build artifacts only.
Client hydration	Only interactive site-local React components hydrate. Noninteractive article/navigation/SEO structure should remain server HTML.
State	Interactive browser state is local to the site/page and never serves as tenant authority.
Metadata	SEO tags/structured data consume DOC-13 SEO projection and DOC-16 rules; Next metadata conveniences may render but not recompute them.
Images	Next Image must use `unoptimized` or a custom fixed-variant loader that selects only DOC-17 `PublicMedia.variants`. Dynamic `/_next/image?w=...` transformation is prohibited.
API routes	Public site apps do not create product/BFF API operations. `/api/v1` remains the shared DOC-15 API routed by trusted deployment infrastructure.
Errors	Local error presentation may be site-specific, but response status, no-store/noindex and wrong-site non-disclosure remain upstream authority.

11. Central Admin Architecture

```
apps/admin/src/
  app/router/ auth/ site-context/ query-client/
  modules/site/{dashboard,posts,categories,tags,authors,media,navigation,seo,redirects,search,calendar,site-
settings}/
  modules/global/{network,websites-domains,admin-users,roles-permissions,audit,platform-settings}/
  access-surfaces/                # login/invite/reset/MFA/reauth; not operational Admin Page IDs
  ui/ styles/ assets/ tests/
```

- Central Admin remains one React/Vite application and preserves exactly 19 modules and 36 operational Admin pages.
- TanStack Query SITE keys begin with explicit site identity plus operation/resource identity. Switching the Active Website invalidates/discards affected state per DOC-10; no browser-global authoritative `currentSite` exists.
- Other browser tabs preserve independent selected-site context. Shared storage cannot grant tenant authority.

- Access surfaces call the 11 API15-AUTH operations and remain outside the 36 operational page count.

12. OpenAPI Source and Frontend Client Generation

- Canonical file: `contracts/http/BLOG-DOC-15_openapi_v1.0-RC7.yaml`. It is the sole executable HTTP source for this baseline.
- `packages/ts/api-contract/generated/` is generated from that YAML and marked DO NOT EDIT. CI regenerates and fails on drift.
- Public client exposes API15-PUB-001/002 only. Admin client exposes API15-AUTH-*, API15-S-* and API15-G-* only.
- Wire fields remain lower_snake_case exactly. Application/domain naming may differ only behind explicit mappers.
- FastAPI route registry is audited operation-by-operation against the canonical YAML: method, path, operation ID, status, params, headers, request/response schema, auth/scope, permissions, rate class, idempotency and concurrency.
- FastAPI's automatically generated `/openapi.json` is a derived diagnostic artifact. It may be disabled in production or compared in CI; it never becomes source authority.

13. Python / FastAPI API Architecture

```
apps/api/
  pyproject.toml                # project metadata, workspace member
  src/bp_api/
    bootstrap/
    http/
      public/                   # API15-PUB-*
      auth/                     # API15-AUTH-*
      admin_site/               # API15-S-*
      admin_global/             # API15-G-*
      dependencies/             # auth/context/request-scoped deps only
      mappers/                  # wire DTO <-> app command/result
    composition/
    health/                     # operational probes, not /api/v1 business API
    config/
```

- FastAPI path functions perform HTTP parsing, dependency injection, Pydantic validation and mapping only. Business rules do not live in route functions.
- Pydantic models are generated/hand-bound to DOC-15 schemas and must use strict extra-field behavior consistent with the API contract. They cannot make an optional field required or silently accept fields not in DOC-15.
- Request-scoped dependency injection constructs authentication principal and SiteContext before protected application use cases. A dependency object is not an authorization shortcut.
- Admin/Preview successful responses preserve no-store and exact DOC-15 headers/statuses. Error mapping uses DOC-15 error codes rather than default FastAPI exception bodies.
- API is stateless for authoritative tenant/session/business state. In-process caches may only hold immutable configuration/compiled schema/provider clients permitted by DOC-20.

14. Python Domain / Application / Ports / Adapters

Layer	Physical home	May depend on	Must not depend on
L23-DOMAIN	packages/python/bp_domain	Python stdlib + pure value libraries	FastAPI, SQLAlchemy, Redis, Celery, OpenSearch, boto3, filesystem/network.
L23-PORTS	packages/python/bp_ports	bp_domain	Concrete providers/frameworks/apps.
L23-APP	packages/python/bp_application	bp_domain + bp_ports	FastAPI/Celery/SQLAlchemy/Redis/ OpenSearch/boto3 concrete clients.
L23-BG-CONTRACT	packages/python/ bp_background_contracts	pure validation/value libraries	frontend/site UI; provider-specific broker internals.
L23-ADAPTERS	packages/python/bp_adapters/*	bp_ports/domain + concrete provider libraries	frontend; reverse domain imports.
L23-ENTRY	apps/api + apps/worker	application + ports + selected adapters	Reusable business logic that exists only in an entrypoint.

DEPENDENCY INVERSION	Domain imports nothing outward. Ports import domain. Application imports domain+ports. Adapters implement ports. API/Worker compose them. Any reverse dependency or framework object leaking into domain/application public interfaces is release-blocking.

15. PostgreSQL Access: SQLAlchemy + Psycopg + RLS

```
database/
  migrations/                    # ordered raw SQL; DOC-08 authority
  seeds/
  rls/
  verification/
packages/python/bp_adapters/postgres/
  engines.py
  site_transaction.py
  platform_transaction.py
  repositories/
  mappings/                      # SQLAlchemy mappings, not schema authority
```

- SQLAlchemy 2.0 is a query/unit-of-work/mapping implementation only. `create_all`, `drop_all`, schema synchronize and application-startup DDL are prohibited in every environment beyond disposable test fixtures that are verified against canonical SQL.
- Psycopg 3 is the PostgreSQL driver. Site-runtime and platform-runtime engine/pool factories use separate credentials and privilege paths.
- Every site-owned repository call occurs inside a transaction that executes `SET LOCAL app.site_id = <trusted site UUID>` before site-owned SQL. Persistent `SET app.site_id` on pooled connections is prohibited.
- `session.execute(text("SET LOCAL app.site_id = :site_id"), ...)` (or equivalent parameter-safe Psycopg execution) is done after BEGIN on the same physical connection; transaction completion clears the setting by PostgreSQL semantics.
- ORM relationship loading must never issue an unscoped site-owned query outside that transaction. Lazy loading after transaction exit is disabled/avoided for site-owned entities.
- DOC-20 pool caps remain: API replica max <=20; Worker process max <=10; aggregate API+Worker+operational configured maxima <=70% of PostgreSQL max_connections.
- Optional read replica remains forbidden for authorization, session validation, Preview, mutations, expected_version, publishing and lifecycle-critical reads.

16. Python Worker and Celery / Redis Broker Mapping

```
apps/worker/
  pyproject.toml
  src/bp_worker/
    bootstrap/
    tasks/
      q18_publish.py
      q18_media.py
      q18_search.py
      q18_sitemap.py
      q18_cache.py
      q18_maintenance.py
    scheduler/
    composition/
    health/
packages/python/bp_background_contracts/
  queue_ids.py job_types.py envelope.py retry.py dedupe.py redis_keys.py
```

DOC-18 logical queue	Celery route	Purpose
q18.publish	q18.publish	Scheduled publication due work.
q18.media	q18.media	Remaining/repair fixed media variants.
q18.search	q18.search	Incremental sync, entity refresh and full site reindex.
q18.sitemap	q18.sitemap	Site-local sitemap generation.
q18.cache	q18.cache	Redis invalidation and required CDN purge.
q18.maintenance	q18.maintenance	Reconciliation/maintenance derived-state work.

The physical Celery task name is a transport mapping to one exact DOC-18 JOB18 ID. There remain exactly six logical source queues and eleven canonical job types. Celery task names, headers or broker message IDs are not business identifiers.

17. Celery Compatibility Controls for DOC-18

IMPORTANT	Celery defaults are not sufficient to freeze this platform. RC2 accepts Celery only with the controls below, because DOC-18 already owns delayed work, deterministic retries, dedupe state, leases, DLQ/replay and current-state fencing.

Concern	Celery/adapter mapping	Locked rule
At-least-once	late acknowledgement + worker-lost reject/redelivery	ACK only after side effect commit or terminal stale/no-op.
Prefetch	worker_prefetch_multiplier=1 for long/idempotent tasks	Prevents one worker from hoarding many tenant jobs; not a fairness authority by itself.
Retry	first-party retry helper computes DOC-18 capped exponential delay + deterministic 0-20% jitter	Disable arbitrary library jitter/default retry policy.
Timeout	per-job soft/hard limits matched to DOC-18 retry profile	Lease/attempt fencing remains final protection against late commit.
Dedupe	DOC-18 control Redis CLAIMED/ENQUEUED/TERMINAL CAS state	Celery task_id uniqueness is not relied on for correctness.
Leases	DOC-18 control Redis lease + attempt token	Celery worker identity does not replace entity/site lease.
DLQ	explicit terminal routing to corresponding dlq18.* physical queue	Expected stale/no-op is ACKed and not dead-lettered.
Site context	validated payload -> trusted CTX-WORKER-SITE -> RLS transaction	Payload site_id is not authorization.
Global coordinator	explicit existing global coordinator path enumerates Site Registry, then emits site jobs	Missing site never means all sites.
Result backend	disabled/ignored unless operationally needed without authority	Task results never become canonical application data.

17.1 Far-Future Scheduled Publishing: Redis/Celery Caveat Resolution

DOC-18 requires immediate delayed acceptance after Schedule/Reschedule and permits far-future schedules, while Celery over a Redis broker warns that ETA/countdown messages can be redelivered when their ETA exceeds the broker visibility timeout. Therefore RC2 does not rely on Celery worker-memory ETA as the durable far-future delay primitive.

1. Producer performs the DOC-18 dedupe CLAIMED transition in control/cache Redis using the canonical idempotency identity.
2. Queue adapter writes the validated JOB18-PUBLISH-DUE envelope into an implementation-private durable sorted-set scheduler area on the dedicated broker Redis plane, keyed by not_before UTC. This is broker plumbing, not an application-visible DOC-18 key and not a seventh logical queue.
3. Only after the broker transaction acknowledges durable storage does producer move dedupe state to ENQUEUED. Ambiguous/failed acceptance leaves the claim recoverable under the DOC-18 RC2 claim lease/state machine.
4. A small scheduler transport loop promotes due envelopes atomically into Celery `q18.publish`. Promotion uses a broker-private claim/ack marker so crash between removal and publish cannot permanently lose work.
5. The mandatory DOC-18 per-site 60-second schedule reconciliation remains active and repairs missing/ambiguous delayed work from PostgreSQL truth. It does not create a new job type.
6. At execution, the worker rechecks site ACTIVE, Post ownership, SCHEDULED state, source_version and scheduled_for_utc. Early/stale delivery never publishes.
7. Library-private broker keys are versioned and confined to the broker Redis credentials. Frontends/API business code never reads them directly; only the queue adapter owns them.

NO DOC-18 CHANGE	This adapter is a PDF-23 queue implementation detail. It preserves immediate delayed enqueue, far-future durability, dedupe retention and 60-second reconciliation without adding a DB table, a product state, a job type or an application-visible Redis key.

18. Redis Control/Cache Architecture

Plane	Client	Authority	Isolation
Control/cache Redis 8.x	redis-py 8.1.x	DOC-18 exact `bp:v1:s:{site_id}:*` / global keys, TTLs, locks, dedupe, cache/search/sitemap coordination.	Site key namespace + trusted context before use.
Queue broker Redis 8.x	Celery/Kombu + queue adapter	Transport durability, source queues, DLQ queues and implementation-private delayed scheduler storage.	Separate endpoint, credentials, persistence, memory policy.
Security-state store	DynamoDB-compatible adapter	DOC-19 logical sessions/MFA/token/step-up/security throttle records.	Not either Redis plane unless DOC-19/18 are explicitly reopened.

- Redis database numbers are not tenant isolation. Site identity is always in the DOC-18 application key grammar.
- Control/cache Redis remains derived/rebuildable and cannot authorize a request or override PostgreSQL truth.
- Broker Redis runs persistence/noeviction settings appropriate for accepted queue work. Control/cache capacity and eviction policy remain DOC-20/24 operational concerns without changing correctness rules.
- No Python code may build DOC-18 Redis keys ad hoc outside the approved key-builder module.

19. OpenSearch Implementation Boundary

- `bp_adapters.search` uses `opensearch-py` against OpenSearch 3.x. Search remains a derived accelerator only.
- Every search document and query includes trusted site scope before provider execution. Network-wide fetch then filter is prohibited.
- DOC-16 normalization, scoring, filtering, eligibility and deterministic tie-break order are reproduced exactly; OpenSearch default BM25 order is never authoritative.
- Full reindex uses site-local shadow generation, validation and atomic activation; incremental work catches up after activation as DOC-18 requires.
- No mandatory second Admin-derived index is created.

20. Media, Object Storage and Image Processing

Concern	Python implementation	Frozen boundary
Object storage	boto3/botocore S3-compatible adapter	Private origin, DOC-17 storage-key grammar, no browser private credentials.
Processing	pyvips/libvips adapter	Orientation, sanitization and exact fixed 16-variant AVIF/WebP catalog only.
Public baseline	DOC-17 `native_webp` generation/verification	Private original has no anonymous fallback; safe baseline must exist before generation switch.
Delivery	CDN/proxy external to app package	Only approved revisioned derivative URLs; no dynamic transform service.
Usage safety	bp_application.media	Authoritative same-site media_usages recheck before replace/delete.

NEXT IMAGE	Next.js image optimization cannot generate arbitrary widths/qualities. Site code uses DOC-17 variants directly through unoptimized/custom-loader behavior. `/_next/image` dynamic transformation is not part of the platform contract.

21. Security Implementation Mapping

DOC-19 area	Python implementation direction	Non-negotiable semantic
Password	argon2-cffi	Exact Argon2id parameters and encoded verifier behavior from DOC-19.
TOTP/recovery	PyOTP + secrets/hashlib/hmac	RFC6238-compatible TOTP and single-use recovery behavior.
Tokens/randomness	Python `secrets` + cryptographic hashing/HMAC	Opaque high-entropy tokens; raw one-time values never persisted/logged.
Sessions/cookies/CSRF	FastAPI/Starlette HTTP adapter + bp_application.auth_security	Opaque server-side session, host-only secure cookie, exact Origin and CSRF rules.
Security state	DynamoDB-compatible conditional/TTL adapter via boto3	Atomic compare/write + TTL; separate from DOC-08 and DOC-18 Redis.
Secrets/envelope	Vault-compatible HTTPS adapter	No secrets in source, Platform Settings, logs or browser bundles.
Mail	SMTP adapter	Invitation/reset delivery only; neutral response behavior.
Audit/privacy	application service + DOC-08 audit_events persistence where authoritative	Event dictionary/privacy minimization unchanged.

The 11 API15-AUTH operations are unchanged. FastAPI must expose exactly those existing transport homes; no extra login/MFA/session operation is created by the Python migration.

22. Observability Implementation Mapping

- OpenTelemetry Python instruments FastAPI, SQLAlchemy/Psycopg, Redis, Celery and OpenSearch adapters. Frontend/server Next.js telemetry may use OpenTelemetry JS only within the Node public runtime.
- Structured JSON logging uses a Python logging adapter (for example structlog or stdlib JSON formatter). Pino is no longer a backend requirement.
- Metrics use Prometheus-compatible Python instrumentation and exact DOC-20 low-cardinality dimensions. Raw slug/email/token/media-private path/content is prohibited as a metric label.
- API spans carry request ID, operation ID and safe context class. Worker spans carry queue, JOB18 type, attempt and safe site identity. Sensitive values are redacted before export.
- Health probes remain operational surfaces outside `/api/v1`; exact deployment wiring remains DOC-24.

23. Testing and Static-Quality Architecture

Area	Tooling	Existing QA authority
TS unit/component	Vitest + Testing Library where needed	Site/Admin package behavior; does not replace QA21 IDs.
Python unit/integration	pytest + pytest-asyncio + testcontainers-python	API/application/adapters/RLS/worker correctness.
Contract	OpenAPI parsers + generated TS types + FastAPI route metadata checker	QA21-API/REG exact DOC-15 parity.
Browser/E2E	Playwright	Admin/public/Preview/multi-tab/site-switch/concurrency.
Accessibility	axe-core + semantic/manual checks	QA21-A11Y.
Performance/DR	k6 + Python/infra runbooks	QA21-PERF/DR + DOC-20 LT scenarios.
TS quality	ESLint + Prettier + dependency-cruiser	Cross-site and frontend import rules.
Python quality	Ruff + mypy + Import Linter + pip-audit	Layer/import/type/security dependency rules.

QA AUTHORITY	DOC-23 defines architecture assertions but does not invent replacement QA21 IDs. Evidence from TS and Python tooling maps to the existing deterministic QA21 release gate and DOC-22 evidence model.

24. Dependency / Import Graph and Enforcement

```
site-* ---> ts/api-client-public ---> ts/api-contract
   |              +--> ts/public-contracts
   +--------------+--> ts/shared-utils

admin ----> ts/api-client-admin ----> ts/api-contract
   +------> ts/shared-utils

api(py) ----> bp_application ---> bp_domain
     |             |              ^
     |             +--> bp_ports -+
     +-----> bp_adapters ---------+
     +-----> bp_observability

worker(py) -> bp_application / bp_background_contracts / bp_adapters / bp_observability

site-A -X-> site-B
TS shared package -X-> apps/sites/* source
Python domain/app -X-> FastAPI/Celery/SQLAlchemy concrete adapters
frontend -X-> DB/Redis/broker/OpenSearch/private storage/security state
```

From	To	Rule
Public site	Own site modules/assets/styles	ALLOWED
Public site	TS public client/contracts/pure utils	ALLOWED
Public site	Another public site / Admin / Python backend	PROHIBITED
Admin	TS admin client/contracts/pure utils	ALLOWED
Admin	Any public site source	PROHIBITED
FastAPI entry	Python application + adapters + OpenAPI metadata	ALLOWED
Worker entry	Python application + background contracts +	ALLOWED

From	To	Rule
	adapters	
API	Worker entrypoint source	PROHIBITED
Worker	API entrypoint source	PROHIBITED
bp_domain	FastAPI/SQLAlchemy/Redis/Celery/OpenSearch/ boto3/apps	PROHIBITED
bp_ports	bp_domain	ALLOWED
bp_application	bp_domain + bp_ports	ALLOWED
bp_adapters	bp_ports + bp_domain + provider libs	ALLOWED
packages/*	apps/sites/* source	PROHIBITED
tests/	production runtime imports	PROHIBITED

Enforcement is mandatory: dependency-cruiser and package export maps for TypeScript/Next/Admin; Import Linter plus module boundaries for Python; CI fails on either graph violation. Generated code is scanned so it cannot hide a site-to-site edge.

25. Naming and Configuration Conventions

Entity	Convention	Example
Repository dirs	kebab-case for app/tooling dirs	site-01-whats-your-rashi
TS packages	@bp/<kebab-case>	@bp/api-client-public
Python packages	snake_case import packages under bp_* namespace	bp_application, bp_adapters.postgres
TS source	kebab-case files; PascalCase React symbols	article-page.tsx / ArticlePage
Python source	snake_case modules; PascalCase classes	site_transaction.py / SiteContext
Functions/vars	camelCase TS; snake_case Python	resolveSiteContext / resolve_site_context
Wire JSON	lower_snake_case	Frozen DOC-15/DOC-13
PostgreSQL	lower_snake_case	Frozen DOC-08 names
API op IDs	API15-*	Exact frozen IDs
Queues/jobs	Q18-* / JOB18-*	Exact frozen IDs
QA IDs	QA21-*	Exact frozen IDs
Env vars	BP_<AREA>_<NAME>	Browser exposure explicitly allowlisted

- Python runtime configuration is parsed once through typed Pydantic Settings or equivalent immutable config object. `os.environ` is not read ad hoc in domain/application modules.
- Node/Next/Admin configuration is parsed once through typed config modules. `process.env` is not read ad hoc in presentation/application code.
- Site/domain registry remains PostgreSQL DOC-08/09 authority, not hard-coded shared config.
- Visual tokens remain site-local and later visual-authority input; runtime config cannot enable an unauthorized feature/permission.

26. Affected Build / Test Orchestration

Change class	Affected execution
One site source/style/asset	Only that Next.js site build/unit tests + repo-wide architecture/security/contract gates.
Shared TS public contract/client	All dependent public sites; Admin only if its client/contract edge is affected.
Admin-only	Admin build/tests + repo-wide gates.
FastAPI transport only	API Python tests/image; Worker unaffected unless shared Python package changed.
Worker bootstrap/task transport only	Worker Python tests/image; API unaffected unless shared Python package changed.
Python domain/application/ports	API + Worker Python test/build nodes.

Change class	Affected execution
Postgres/Redis/search/media adapter	API and/or Worker according to explicit import graph; integration tests required.
Canonical OpenAPI	Contract diff + TS codegen + API contract tests + all actual client dependents.
DOC-08 migration/seed	DB verification + API/Worker integration + all QA stages required by DOC-21/DOC-24.
Root pnpm/uv lock or repo policy	Repository-wide dependency/security checks and all affected app/runtime builds.

Turborepo is the top-level graph and invokes Python tasks through bridge scripts such as `pnpm turbo run test --filter=@bp/api-runtime` -> `uv run pytest ...`. The bridge does not make Node.js the API runtime. Python dependency changes under `packages/python/**` invalidate API/Worker task caches through declared inputs.

27. Local Development and Future-Site Scaffold

- Local Compose can start PostgreSQL, control/cache Redis, broker Redis, OpenSearch, S3-compatible storage, security-state emulator, Vault-compatible dev adapter and test SMTP independently of all applications.
- A developer may start `site-03 + API + required dependencies` without running the other 13 site servers or Admin/Worker unless the task needs them.
- API/Worker local commands use `uv run`; site/Admin use `pnpm --filter`. No production credentials are permitted locally.
- Database startup/migrate tooling executes canonical raw SQL; API/Worker startup never mutates schema.
- Future-site generator copies a neutral Next.js scaffold to `apps/sites/site-15-...`, creates all frozen public Page-ID implementation homes and local shell/assets, then stops. It does not create DB site/domain rows, permissions, aliases, content or deployment resources.
- Shared backend/API/Worker discovers runtime sites through the authoritative Site Registry, not through a SITE enum requiring code edits.

28. Lockfile and Upgrade Policy

Artifact	Authority rule
pnpm-lock.yaml	Exact JS/TS resolved graph. CI uses frozen lockfile; no install may mutate it.
uv.lock	Exact Python resolved graph. CI uses locked/frozen resolution; no unreviewed transitive drift.
package.json / pnpm catalog	Human-readable allowed major/minor ranges and workspace edges; exact versions still lockfile-owned.
pyproject.toml	Python direct dependency constraints, workspace membership and tool config; exact versions still uv.lock-owned.
Container base digest	Deployment artifact authority later DOC-24; semantic tags are not immutable release identity.

Change	Required review
Patch within frozen line	Dependency/security scan + affected unit/integration + contract checks; update lockfile in reviewed PR.
Minor within approved major where no contract change	Compatibility review + affected QA21 + performance/security regression where material.
Major Next.js/Python/FastAPI/SQLAlchemy/PostgreSQL/Redis/OpenSearch/Celery change	Controlled DOC-23 impact review before adoption.
Queue/provider behavior change	Prove all DOC-18 scheduling/dedupe/retry/lease/DLQ semantics; otherwise blocker/reopen.
Schema/API/product semantic change	Owned upstream document must be revised first; DOC-23 cannot authorize it.

29. Complete PDF-01 through PDF-22 Cross-Document Impact Matrix

The audit searched the available authority set for Node.js, Fastify, Kysely, node-postgres, ioredis, BullMQ, Astro, Node worker/security/mail/observability/test references and framework-specific public-site assumptions. Those concrete implementation choices occur in DOC-23 RC1 (and the downstream DOC-24 RC1), while PDFs 01-22 remain semantic authorities. Where an earlier document explicitly delegates implementation technology to DOC-23, RC2 changes only the implementation mapping.

PDF	Existing decision/reference	Impact of revised stack	Classification	Required action	Owner
PDF-01	Platform topology: 14 independent public frontends + Central Admin/shared backend/data core.	Next.js/Python preserve topology; no product behavior changes.	NO CHANGE REQUIRED	PDF 01 remains frozen; only PDF-23 implementation alignment is required.	DOC-01
PDF-02	Roles/permissions/ authorization and implementation-neutral pseudocode.	Python auth/application services implement identical order/outcomes.	NO CHANGE REQUIRED	PDF 02 remains frozen; only PDF-23 implementation alignment is required.	DOC-02
PDF-03	161 capabilities; visual frontend autonomy; implementation mechanisms delegated later.	Framework/language does not alter capability existence/scope.	NO CHANGE REQUIRED	PDF 03 remains frozen; only PDF-23 implementation alignment is required.	DOC-03
PDF-04	Trusted SiteContext, context registry, isolation and RLS direction.	FastAPI/Celery explicitly reconstruct same contexts; no new context.	NO CHANGE REQUIRED	PDF 04 remains frozen; only PDF-23 implementation alignment is required.	DOC-04
PDF-05	14 public Page IDs / route inventory.	Next.js adapters implement same Page IDs; no route family added.	NO CHANGE REQUIRED	PDF 05 remains frozen; only PDF-23 implementation alignment is required.	DOC-05
PDF-06	19 Admin modules / 36 operational pages.	React/Vite Admin unchanged.	NO CHANGE REQUIRED	PDF 06 remains frozen; only PDF-23 implementation alignment is required.	DOC-06

29. Complete PDF-01 through PDF-22 Cross-Document Impact Matrix - Continued 2/4

PDF	Existing decision/reference	Impact of revised stack	Classification	Required action	Owner
PDF-07	Post/content/editorial lifecycle, versions, scheduling, autosave/revisions.	Python services and worker implement same transitions.	NO CHANGE REQUIRED	PDF 07 remains frozen; only PDF-23 implementation alignment is required.	DOC-07
PDF-08	33-table PostgreSQL model, 20 site-owned/RLS; explicitly delegates repository/ORM choice to PDF-23.	Kysely/node-postgres -> SQLAlchemy/Psycopg only. Schema/RLS/raw SQL semantics unchanged.	NO CHANGE REQUIRED	PDF 08 remains frozen; only PDF-23 implementation alignment is required.	DOC-08
PDF-09	Domain/URL/slug/routing/ Site resolution.	Next.js/FastAPI preserve host and exact route outcomes; no framework route authority.	NO CHANGE REQUIRED	PDF 09 remains frozen; only PDF-23 implementation alignment is required.	DOC-09
PDF-10	36 Admin page flows/states, site switching, actions.	Admin stack already React/Vite; backend transport implementation changes only.	NO CHANGE REQUIRED	PDF 10 remains frozen; only PDF-23 implementation alignment is required.	DOC-10
PDF-11	14 public page flows/states/status behavior.	Next.js Pages Router SSR maps exact 200/redirect/404/410/Previe w semantics.	NO CHANGE REQUIRED	PDF 11 remains frozen; only PDF-23 implementation alignment is required.	DOC-11
PDF-12	Semantic public component/presentation contracts; site-local authority.	React components implement same semantics independently per site.	NO CHANGE REQUIRED	PDF 12 remains frozen; only PDF-23 implementation alignment is required.	DOC-12

29. Complete PDF-01 through PDF-22 Cross-Document Impact Matrix - Continued 3/4

PDF	Existing decision/reference	Impact of revised stack	Classification	Required action	Owner
PDF-13	Canonical PageData/public data contracts.	Generated TS clients + FastAPI mappers preserve fields/order/nullability; no new projection.	NO CHANGE REQUIRED	PDF 13 remains frozen; only PDF-23 implementation alignment is required.	DOC-13
PDF-14	14-application frontend model; site owns shell/UI/styles/assets; cross-site imports forbidden.	Astro was not DOC-14 authority; Next.js is a valid implementation of the same boundary.	NO CHANGE REQUIRED	PDF 14 remains frozen; only PDF-23 implementation alignment is required.	DOC-14
PDF-15	RC7 HTTP/OpenAPI: 112 operations/85 paths/217 schemas; exact auth/scope/status/idempotency/concurrency.	FastAPI implements, never regenerates/redefines, canonical YAML.	NO CHANGE REQUIRED	PDF 15 remains frozen; only PDF-23 implementation alignment is required.	DOC-15
PDF-16	SEO/search/sitemap/ canonical/redirect algorithms.	OpenSearch Python adapter and Next SSR render exact outputs; no algorithm change.	NO CHANGE REQUIRED	PDF 16 remains frozen; only PDF-23 implementation alignment is required.	DOC-16
PDF-17	S3/private original/native_webp/fixed variants/sanitization/CDN.	boto3 + pyvips replace JS SDK/Sharp implementation without external behavior change.	NO CHANGE REQUIRED	PDF 17 remains frozen; only PDF-23 implementation alignment is required.	DOC-17
PDF-18	6 queues, 11 jobs, delayed scheduling, dedupe/retries/leases/DLQ/ cache; queue provider/language explicitly delegated to PDF-23.	Celery+Redis adapter must preserve every logical rule; far-future delay handled in broker adapter.	NO CHANGE REQUIRED	PDF 18 remains frozen; only PDF-23 implementation alignment is required.	DOC-18

29. Complete PDF-01 through PDF-22 Cross-Document Impact Matrix - Continued 4/4

PDF	Existing decision/reference	Impact of revised stack	Classification	Required action	Owner
PDF-19	Auth/session/MFA/CSRF/ CORS/step-up/secrets/ audit/privacy/recovery mechanics.	Python-compatible cryptography/session/provi der adapters implement same policy; AUTH transport unchanged.	NO CHANGE REQUIRED	PDF 19 remains frozen; only PDF-23 implementation alignment is required.	DOC-19
PDF-20	Performance/scale/pools/ concurrency/observability/ backup/DR targets.	FastAPI/Celery/Next must meet same budgets; OTel Python replaces Node instrumentation for backend.	NO CHANGE REQUIRED	PDF 20 remains frozen; benchmark evidence is required before implementation verification.	DOC-20
PDF-21	180 deterministic release-blocking QA IDs and evidence rules.	Python test tooling added; Playwright/axe/k6 remain; expected behavior unchanged.	NO CHANGE REQUIRED	PDF 21 remains frozen; harness implementation follows revised DOC-23.	DOC-21
PDF-22	Master traceability/change propagation/evidence states.	Technology home references point to DOC-23 RC2 at implementation time; product trace graph unchanged.	NO CHANGE REQUIRED	PDF 22 remains frozen; implementation remains NOT_VERIFIED/EVIDENC E_REQUIRED.	DOC-22

29.1 Technology-Specific Search Result

Term family searched	Result within PDFs 01-22
Node.js / Fastify / Kysely / node-postgres	No canonical implementation mandate found in PDFs 01-22; concrete baseline was DOC-23 RC1.
Astro / Astro SSR adapter	No upstream framework mandate found; concrete baseline was DOC-23 RC1.
BullMQ / JavaScript worker	DOC-18 explicitly delegates queue provider/language to DOC-23; BullMQ was DOC-23 RC1 implementation.
ioredis / Node crypto / Nodemailer / Pino	No upstream semantic authority depends on these libraries; they were DOC-23 implementation choices.
Node-specific test/package conventions	QA semantics remain DOC-21; repository/package conventions are DOC-23 authority.

30. Required Downstream Changes and Explicit Lists

30.1 Downstream document requiring revision

PDF-24 REQUIRES REVISION	PDF 24 requires revision because its RC1 deployment contract explicitly defines Node 24 Astro SSR site images, Node/Fastify API image, Node Worker image, Node migration image and pnpm-only build/install assumptions. Those deployment mechanisms cannot remain canonical after DOC-23 RC2 changes the public sites to Next.js and backend/worker to Python.

- Preserve DOC-24 semantic release rules: four environments, immutable build-once promotion, selective site deployment, Docker/YAML conventions, QA gates, migration safety, health/readiness, rollback and production safeguards.
- Replace IMG24-SITE-* runtime description with Node 24 + Next.js 16.3 SSR; keep one image/deployment per affected site.
- Replace IMG24-API and IMG24-WORKER with Python 3.13 images using uv.lock/frozen Python dependencies; Worker profile topology remains the same six logical queue profiles plus scheduler deployment path.
- Replace migration image implementation with Python/psql tooling or a minimal migration image capable of executing canonical raw SQL; no ORM migration authority.
- Update CI/CD from one pnpm lockfile assumption to two ecosystem lockfiles and mixed JS/Python checks. Keep immutable digest promotion and affected graph rules.
- Keep existing AWS/Kubernetes/provider topology unless a provider incompatibility is independently found; language change alone does not require topology redesign.

30.2 PDFs requiring NO semantic change

PDFs 01 through 22: all remain frozen at their existing semantic authority. The technology migration is absorbed by PDF-23 RC2 because those documents either define implementation-neutral behavior or explicitly delegate concrete engineering choices to PDF-23.

30.3 PDFs requiring implementation-reference updates

Within PDFs 01-22, no mandatory Node/Fastify/Kysely/Astro/BullMQ implementation reference was found that requires rewriting the document. The explicit implementation-reference update is downstream PDF-24 RC1. If an authoritative copy of any PDF 01-22 later reveals a technology-specific non-semantic note not present in the audited set, classify it as IMPLEMENTATION NOTE ONLY and update only that note under the document's change-control process; do not reopen its semantic rules.

31. Genuine Conflicts / Blockers

Candidate issue	Finding	Disposition
Next.js exact HTTP status control	No blocker when canonical public routes use Pages Router SSR and explicit response status/redirect handling.	RESOLVED IN DOC-23 RC2
Next.js image optimization vs fixed DOC-17 variants	Built-in arbitrary dynamic transformation would conflict.	RESOLVED: prohibit dynamic optimizer; use fixed variant URLs.
FastAPI auto-OpenAPI vs DOC-15 authority	Framework-generated schema could drift.	RESOLVED: DOC-15 YAML remains source; CI parity check; generated schema is derived only.
SQLAlchemy schema metadata vs DOC-08	ORM schema push would create second authority.	RESOLVED: raw SQL only; create_all/sync prohibited.
Celery Redis far-future ETA	Pure Celery ETA/visibility-timeout behavior is not sufficient for DOC-18 far-future durable delayed enqueue.	RESOLVED by broker-private durable delayed adapter + 60s reconciliation; no new logical queue/job/key authority.
Celery retry/dedupe/DLQ defaults	Library defaults do not match exact DOC-18 semantics.	RESOLVED by first-party background-contract helpers and explicit routing/fencing.
Python language change and DOC-15	No wire-contract impact.	NO BLOCKER
Python language change and DOC-20 SLOs	Feasibility requires later runtime/load evidence, not specification redesign.	NO SPEC BLOCKER; EVIDENCE REQUIRED

BLOCKER COUNT	No unresolved genuine blocker remains inside PDF-23 RC2 after the explicit Celery/Redis delayed-work adapter rule. However, PDF-24 RC1 is incompatible as a deployment implementation document and must be revised before the documentation chain can be considered globally synchronized.

32. Implementation Evidence and Status Governance

Item	Current status	Evidence required
Specification mapping	SPEC_MAPPED	This RC2 defines required physical architecture and upstream/downstream impact.
Repository implementation	NOT_VERIFIED	Candidate Git tree, manifests, pyproject, lockfiles, import graph.
Next.js site independence	EVIDENCE_REQUIRED	Independent build logs + dependency scan + representative one-site change.
FastAPI/OpenAPI parity	EVIDENCE_REQUIRED	112-operation method/path/status/schema/security comparison to RC7 YAML.
DB/RLS implementation	EVIDENCE_REQUIRED	Raw migration + SQLAlchemy/Psycopg transaction tests + 33/20/RLS checks.
Celery/DOC-18 parity	EVIDENCE_REQUIRED	6/11 registry, delayed durability, retry/dedupe/lease/DLQ/reconcile tests.
Performance	EVIDENCE_REQUIRED	DOC-20 LT20 and QA21-PERF evidence against candidate stack.
Runtime QA	NOT_EXECUTED	DOC-21 results tied to same commit/build.

33. Traceability

Trace ID	Upstream	RC2 implementation lock
TR23-001	DOC-01	Platform topology -> 14 Next.js site apps + one Admin + Python API/Worker in one Hybrid Monorepo.
TR23-002	DOC-02	Authorization -> server-side Python application/security boundaries; no UI authorization shortcut.
TR23-003	DOC-03	Feature catalog -> code folders without new

Trace ID	Upstream	RC2 implementation lock
		capabilities.
TR23-004	DOC-04	SiteContext/isolation -> explicit request/job context + RLS transaction.
TR23-005	DOC-05	Public Page IDs -> per-site Next.js route/page-module homes.
TR23-006	DOC-06	Admin IA -> one React/Vite Admin, 19 modules/36 pages.
TR23-007	DOC-07	Content workflow -> Python domain/application use cases and DOC-18 worker effects.
TR23-008	DOC-08	DB schema/RLS -> raw SQL + SQLAlchemy/Psycopg access + SET LOCAL.
TR23-009	DOC-09	Routes -> Next.js SSR adapters preserve Host and exact status/Location.
TR23-010	DOC-10	Admin flows -> module/page-local queries/mutations/state.
TR23-011	DOC-11	Public flows -> SSR status/state/Preview/failure handling.
TR23-012	DOC-12	Component semantics -> site-local React presentation.
TR23-013	DOC-13	PageData -> generated TS types + FastAPI wire mappers.
TR23-014	DOC-14	Frontend model -> physical no-cross-site import + local shell/assets.
TR23-015	DOC-15 RC7	HTTP/OpenAPI -> canonical YAML + FastAPI parity + generated clients.
TR23-016	DOC-16 RC4	SEO/search -> server-rendered SEO + OpenSearch adapter exact algorithm.
TR23-017	DOC-17 RC2	Media -> boto3/pyvips fixed media adapters.
TR23-018	DOC-18 RC2	Workers/cache -> Python background contracts + Celery/Redis adapter preserving 6/11 and delayed semantics.
TR23-019	DOC-19 RC2	Security -> Python crypto/session/security-state/Vault/mail adapters.
TR23-020	DOC-20 RC1	Scale/DR/observability -> stateless FastAPI/Next processes, DB pool caps, OTel evidence.
TR23-021	DOC-21 RC1	QA -> mixed JS/Python tests map to existing QA21 IDs.
TR23-022	DOC-22 RC1	Trace/evidence -> NOT_VERIFIED state and change-propagation linkage.

34. Comprehensive Acceptance Criteria

ID	Release-blocking architecture assertion
AC23-001	One Git Hybrid Monorepo contains exactly the initial 14 site app packages + one Admin + one API + one Worker application class, without a lifetime hard-code to 14.
AC23-002	Every public site is a separate Next.js package that can build/test/deploy without building another site when no shared dependency changed.
AC23-003	All 14 site apps use React + TypeScript as primary public UI implementation and retain site-local shell, Header, Footer, navigation, cards, buttons, styles, assets and presentation.
AC23-004	No site imports another site by source path, workspace dependency, alias, re-export, symlink or generated module.
AC23-005	No shared package owns public visual components/theme/assets or mutable current-site state.
AC23-006	Every frozen public Page ID has a route/page-module implementation home

ID	Release-blocking architecture assertion
	inside every site app.
AC23-007	Next.js public routes preserve exact DOC-09/11 200/301/302/308/404/410/Preview outcomes.
AC23-008	Unknown Host never falls back to the site whose deployment happened to receive traffic.
AC23-009	Internal public links consume resolved PublicDestination URLs and do not reconstruct canonical URLs from slugs.
AC23-010	Authoritative PageData fetches do not use a Next.js semantic cache that can outrank DOC-18.
AC23-011	Next.js dynamic image transformation is disabled/bypassed for CMS media; only DOC-17 fixed variants are emitted.
AC23-012	No site-local Next.js API route creates a second business API contract.
AC23-013	Central Admin remains one React/Vite app with exactly 19 modules and 36 operational pages.
AC23-014	Admin SITE query keys and state include explicit site context; site switching cannot let stale responses overwrite new context.
AC23-015	Admin imports no public site presentation source.
AC23-016	Python 3.13 is the only backend/worker runtime baseline; Node remains frontend/build/orchestration only.
AC23-017	FastAPI exposes the exact DOC-15 RC7 operation registry without extra product operations.
AC23-018	Canonical DOC-15 RC7 YAML exists once and remains source-of-truth over FastAPI generated OpenAPI.
AC23-019	CI compares FastAPI method/path/operationId/status/schema/security/idempotency/concurrency/rate metadata against the canonical YAML.
AC23-020	Pydantic validation does not broaden/rename wire fields or weaken unknown-field rules.
AC23-021	HTTP route functions contain no canonical lifecycle/permission/search/media/storage/Redis business logic.
AC23-022	Python domain imports no FastAPI, SQLAlchemy, Redis, Celery, OpenSearch or boto3 adapter.
AC23-023	Python ports depend inward on domain; application depends on domain+ports; adapters implement ports; entrypoints compose.
AC23-024	API and Worker do not import each other's entrypoint source.
AC23-025	FastAPI request dependencies cannot replace server-side authorization/use-case checks.
AC23-026	PostgreSQL remains version 17.x and DOC-08 remains semantic/DDL authority.
AC23-027	Exactly DOC-08 33 canonical tables and 20 site-owned RLS tables are implemented; SQLAlchemy adds none.
AC23-028	All schema evolution occurs through ordered raw SQL migrations; create_all/schema push/synchronize is disabled.

ID	Release-blocking architecture assertion
AC23-029	Site transactions execute BEGIN -> SET LOCAL app.site_id from trusted context -> scoped work -> COMMIT/ROLLBACK on one connection.
AC23-030	No pooled connection retains tenant state after transaction return.
AC23-031	Site-runtime and platform-runtime engines/pools/credentials remain separate.
AC23-032	Lazy/deferred ORM loading cannot escape the tenant-scoped transaction for site-owned records.
AC23-033	API pool max <=20 and Worker pool max <=10; aggregate configured maxima <=70% max_connections.
AC23-034	Read replicas, if enabled, are excluded from authorization/session/Preview/mutation/version/lifecycle-critical reads.

ID	Release-blocking architecture assertion
AC23-035	Worker implementation exposes exactly six DOC-18 logical queues and eleven canonical JOB18 job types.
AC23-036	Celery task names/IDs do not become new product/job identifiers.
AC23-037	Delivery is treated as at-least-once; ACK happens only after side effect commit or terminal stale/no-op.
AC23-038	Every site-owned job reconstructs trusted CTX-WORKER-SITE and enters RLS before target access.
AC23-039	Missing site context never means all sites; global coordinator only enumerates sites then emits site-scoped work.
AC23-040	DOC-18 deterministic retry/backoff/jitter/timeout profiles are implemented by first-party helpers rather than Celery defaults.
AC23-041	DOC-18 dedupe CLAIMED/ENQUEUED/TERMINAL state remains in control Redis and Celery uniqueness is not correctness authority.
AC23-042	DOC-18 leases and attempt tokens fence stale workers; broker worker identity is insufficient.
AC23-043	Each source queue has the DOC-18 corresponding DLQ/replay behavior; stale/no-op work does not enter DLQ.
AC23-044	Far-future publish work is durably parked in implementation-private broker storage and not in Celery worker-memory ETA alone.
AC23-045	Delayed broker acceptance moves dedupe to ENQUEUED only after positive durable acknowledgement.
AC23-046	Delayed promotion is crash-safe and the 60-second site schedule reconciliation remains active.
AC23-047	Reschedule/version changes make old due messages stale without new DB fields.
AC23-048	Broker Redis is physically/logically separate from control/cache Redis.
AC23-049	Celery result backend/output never becomes canonical application state.
AC23-050	Control/cache Redis uses exact DOC-18 site/global key builders and TTL semantics; no ad-hoc key grammar.
AC23-051	Redis DB number is not tenant isolation; site identity is explicit in key and context.
AC23-052	OpenSearch is derived-only and every document/query is site scoped before provider execution.
AC23-053	OpenSearch default BM25 does not determine canonical result order; golden fixtures prove DOC-16 ranking/tie-breaks.
AC23-054	Full reindex preserves shadow generation/validation/atomic site activation and incremental catch-up.
AC23-055	S3-compatible object storage uses private origin and DOC-17 exact key/delivery rules.
AC23-056	pyvips emits only DOC-17 fixed derivatives and strips metadata exactly as required.

ID	Release-blocking architecture assertion
AC23-057	`native_webp` is verified before safe public generation switch; private original is never anonymous fallback.
AC23-058	No arbitrary media transform endpoint or URL width/quality API is introduced.
AC23-059	Argon2id implementation matches DOC-19 parameters exactly.
AC23-060	TOTP/recovery/session/CSRF/CORS/step-up behavior matches DOC-19 and the 11 AUTH operations remain unchanged.
AC23-061	Security-state records remain outside DOC-18 Redis and DOC-08 content tables unless upstream is reopened.
AC23-062	Secrets/tokens/credentials are absent from source, browser bundles, logs, queues and PageData.
AC23-063	Invitation/reset raw token exists only within the approved one-time delivery boundary.

ID	Release-blocking architecture assertion
AC23-064	`pnpm-lock.yaml` is the sole JS/TS dependency lock; package-local JS lockfiles are absent.
AC23-065	`uv.lock` is the sole Python dependency lock; package-local Python locks/floating requirements are absent.
AC23-066	CI uses frozen/locked install for both ecosystems and fails on lock drift.
AC23-067	Turborepo can calculate affected frontend and Python task nodes through explicit bridge manifests/inputs.
AC23-068	A Python-core change affects API+Worker; an API-entry-only change does not force Worker; one-site change does not force other sites.
AC23-069	dependency-cruiser enforces JS/TS edges and Import Linter enforces Python layer edges.
AC23-070	Generated code is included in dependency scans and cannot create hidden site-to-site imports.
AC23-071	Ruff/mypy/ESLint/TypeScript checks pass for the candidate commit.
AC23-072	OpenAPI TS generated output is reproducible from canonical YAML and drift-checked.
AC23-073	Local Compose can run one selected site + API + required dependencies without the other 13 public apps.
AC23-074	Local/test credentials are non-production and no production secret appears in repository files.
AC23-075	API/Worker startup performs no schema-changing DDL.
AC23-076	Future-site scaffold copies neutral Next.js structure and has no runtime dependency on scaffold source.
AC23-077	Future-site generator creates no DB site/domain row, permission, alias, content or deployment resource.
AC23-078	Adding future site does not require shared backend SITE switch statements.
AC23-079	PDFs 01-22 remain semantically unchanged and the complete impact matrix has no CONTRACT IMPACT or GENUINE CONFLICT classification.
AC23-080	DOC-08 ORM delegation remains intact; SQLAlchemy/Psycopg do not become DB semantic authority.
AC23-081	DOC-14 frontend ownership remains intact; Next.js does not create shared visual runtime.
AC23-082	DOC-15 RC7 OpenAPI remains unchanged by FastAPI implementation.
AC23-083	DOC-17 media contract remains unchanged by boto3/pyvips adapters.
AC23-084	DOC-18 logical worker/cache contract remains unchanged by Celery/Redis mapping.

ID	Release-blocking architecture assertion
AC23-085	DOC-19 security contract remains unchanged by Python library mapping.
AC23-086	DOC-20 SLO/pool/concurrency/DR/observability requirements remain acceptance targets for the new stack.
AC23-087	DOC-21 deterministic QA IDs/evidence remain release authority; Python tooling maps into them rather than replacing them.
AC23-088	DOC-22 implementation status stays NOT_VERIFIED/EVIDENCE_REQUIRED until candidate evidence exists.
AC23-089	DOC-24 RC1 is flagged for controlled implementation revision before global documentation-chain synchronization.
AC23-090	No new product feature, permission, capability, Admin page, public Page ID, DB object, API operation, Post status, media variant, job type or context is introduced.
AC23-091	No superseded Astro/Fastify/Kysely/node-postgres/ioredis/BullMQ assumption remains normative inside DOC-23 RC2.
AC23-092	Major runtime/framework/database/search/Redis/queue upgrades trigger controlled DOC-23 impact review.
AC23-093	All applicable AC23 assertions are mapped to existing QA21/DOC-22 evidence

ID	Release-blocking architecture assertion
	classes for implementation verification.

35. Freeze-Readiness Checklist and Final Verdict

Freeze ID	Required condition
FZ23-01	Hybrid Monorepo with 14 independent Next.js sites + Admin + Python API + Python Worker accepted.
FZ23-02	Exact 14 site package/folder identities and future-site non-ceiling strategy accepted.
FZ23-03	Next.js 16.3.x / React 19 / TypeScript public baseline accepted.
FZ23-04	React/Vite/Router/TanStack Central Admin baseline accepted.
FZ23-05	Python 3.13 + FastAPI/Pydantic API baseline accepted with DOC-15 RC7 supremacy.
FZ23-06	PostgreSQL 17 + SQLAlchemy 2.0 + Psycopg 3 access mapping accepted; raw SQL remains authority.
FZ23-07	Redis 8 control/cache + separate Redis 8 broker planes accepted.
FZ23-08	Celery 5.6 execution mapping and explicit far-future delayed broker adapter accepted.
FZ23-09	Exact DOC-18 6 queues/11 jobs/retry/dedupe/lease/DLQ/reconcile semantics preserved.
FZ23-10	OpenSearch 3.x Python adapter and deterministic DOC-16 ranking boundary accepted.
FZ23-11	S3/boto3 + pyvips fixed-media adapter accepted; Next image optimizer dynamic transforms prohibited.
FZ23-12	Python DOC-19 security adapter mapping accepted; no security-state move into Redis.
FZ23-13	Mixed pnpm + uv workspace and two authoritative lockfiles accepted.
FZ23-14	Turborepo top-level affected graph + Python bridge inputs accepted.
FZ23-15	dependency-cruiser + Import Linter enforcement accepted.
FZ23-16	Public site-local ownership/no shared visual runtime accepted.
FZ23-17	DOC-01..22 impact matrix has no unresolved semantic conflict.
FZ23-18	No superseded Node-backend/Astro/Fastify/Kysely/BullMQ assumption remains normative in RC2.
FZ23-19	DOC-24 RC1 downstream revision requirement recorded and cannot be ignored before global chain freeze.
FZ23-20	Implementation remains NOT_VERIFIED/EVIDENCE_REQUIRED until repository/build/QA evidence exists.
FZ23-21	Current technology lines mutually compatible at review date; exact patches delegated to lockfiles.
FZ23-22	93 AC23 acceptance assertions accepted as revised architecture baseline.

FINAL VERDICT	DOC-23 v1.0-RC2 is internally coherent and FREEZE-READY as the revised engineering-architecture authority, subject to project-owner approval. PDFs 01-22 remain aligned and require no semantic revision. No unresolved genuine blocker exists inside the revised stack. PDF-24 RC1 requires a controlled revision because its Docker/CI/runtime implementation still targets Astro/Fastify/Node Worker/pnpm-only assumptions. Global documentation-chain synchronization is therefore pending DOC-24 revision.

Appendix A - Complete Revised Repository Tree

```
/
├─ package.json
├─ pnpm-workspace.yaml
├─ pnpm-lock.yaml
├─ turbo.json
├─ tsconfig.base.json
├─ eslint.config.mjs
├─ prettier.config.mjs
├─ dependency-cruiser.config.cjs
├─ pyproject.toml
├─ uv.lock
├─ ruff.toml
├─ mypy.ini
├─ importlinter.ini
├─ apps/
│  ├─ sites/
│  │  ├─ site-01-whats-your-rashi/ ... site-14-vastu-shastra-360/
│  ├─ admin/
│  ├─ api/
│  │  ├─ package.json             # Turborepo bridge only
│  │  ├─ pyproject.toml
│  │  └─ src/bp_api/...
│  └─ worker/
│     ├─ package.json             # Turborepo bridge only
│     ├─ pyproject.toml
│     └─ src/bp_worker/...
├─ packages/
│  ├─ ts/
│  │  ├─ api-contract/
│  │  ├─ api-client-public/
│  │  ├─ api-client-admin/
│  │  ├─ public-contracts/
│  │  ├─ shared-utils/
│  │  └─ tooling-config/
│  └─ python/
│     ├─ bp-domain/
│     ├─ bp-ports/
│     ├─ bp-application/
│     ├─ bp-background-contracts/
│     ├─ bp-adapters-postgres/
│     ├─ bp-adapters-redis/
│     ├─ bp-adapters-queue/
│     ├─ bp-adapters-search/
│     ├─ bp-adapters-object-storage/
│     ├─ bp-adapters-media/
│     ├─ bp-adapters-security-store/
│     ├─ bp-adapters-secrets/
│     ├─ bp-adapters-mail/
│     ├─ bp-observability/
│     └─ bp-test-support/
├─ contracts/http/BLOG-DOC-15_openapi_v1.0-RC7.yaml
├─ database/{migrations,seeds,rls,verification}/
├─ infrastructure/{local,images,templates,observability}/
├─ tests/{contract,isolation,dependency,e2e,accessibility,performance,dr,fixtures}/
├─ tooling/{generators,scaffolds/public-site,scripts,dependency-rules,configs}/
└─ docs/{authorities,adr}/
```

Appendix B - Package / Import Dependency Matrix

Package/zone	May depend on	May be consumed by	Must never contain/import
ts/api-contract	No runtime app source	Frontend clients/apps as typed contract source	UI, site state, invented fields.
ts/api-client-public	api-contract, pure utils	site-*	Admin ops, site singleton, backend internals.
ts/api-client-admin	api-contract, pure utils	admin	Public site UI, backend internals.
ts/public-contracts	api-contract where required	site-*	Visual components/theme/assets.

Package/zone	May depend on	May be consumed by	Must never contain/import
ts/shared-utils	pure libraries	approved TS consumers	I/O, mutable tenant state, site constants.
bp_domain	pure Python libs	ports/application/adapters via exports	FastAPI, SQLAlchemy, Redis, Celery, OpenSearch, boto3, apps.
bp_ports	bp_domain	application/adapters	concrete providers/apps.
bp_application	domain + ports	api/worker	concrete adapters at import time.
bp_background_contracts	pure validation/value libs	api/worker/queue/redis adapter where needed	frontend UI/provider-specific broker internals.
bp_adapters.postgres	ports/domain + SQLAlchemy/Psycopg	api/worker composition	frontend/schema authority.
bp_adapters.redis	ports/background + redis-py	api/worker	frontend/security state.
bp_adapters.queue	ports/background + Celery/Kombu/redis client	api/worker	frontend/business semantics.
bp_adapters.search	ports + opensearch-py	api/worker	frontend/default ranking authority.
bp_adapters.object_storage	ports + boto3	api/worker	frontend private-origin access.
bp_adapters.media	ports + pyvips	api/worker	frontend/dynamic transforms.
bp_adapters.security_store	ports + conditional TTL client	api	DOC-18 Redis.
bp_observability	OTel/logging/metrics libs	api/worker/server adapters	business decisions/site presentation.

Appendix C - Per-Site Next.js Page-Local Template

```
src/site/pages/article/
  article-page.tsx               # site-local React composition
  article-page.model.ts          # DOC-13 PageData -> local view model
  article-page.module.css        # site-local presentation
  components/
    article-header.tsx
    article-body-renderer.tsx
    related-articles.tsx
  tests/article-page.spec.tsx

src/pages/article/[post_slug].tsx
  # thin SSR route adapter only; fetches canonical API and maps exact status
```

- May import same-site shell/components and approved TS public packages.
- May not import another site, Admin, Python backend or infrastructure adapter.
- May not add PageData fields or change ArticleBody semantics.
- Visual code stays page-local unless genuinely reused by multiple pages of the same site.
- The same locality rule applies to Home, Category, Tag, Author, Search, five Static pages, Preview, Not Found and Gone.

Appendix D - Python Backend Domain / Layer Map

Responsibility	Physical home	Boundary
HTTP public/auth/site/global	apps/api/src/bp_api/http/*	Maps exact API15 operations; no business logic.
Public PageData	bp_application/public_page	Calls route/site/data/SEO/media/search ports; yields DOC-13-compatible result.
SITE Admin use cases	bp_application/site/*	Posts/taxonomy/authors/media/navigation/SEO/redirect/search/calendar/settings.
GLOBAL Admin use cases	bp_application/global/*	Network/sites-domains/admin-users/roles/audit/platform-settings.
Security use cases	bp_application/auth_security	DOC-19 login/MFA/session/step-up/Preview mechanics.
Background use cases	bp_application/background	Publish/media/search/sitemap/cache/maintenance functions invoked by Worker.
Repositories	bp_adapters/postgres	Exact DOC-08 rows/relations under trusted transaction wrappers.
Derived adapters	bp_adapters/{redis,search,queue,media,object_storage}	DOC-16/17/18 derived systems only.
Security providers	bp_adapters/{security_store,secrets,mail}	DOC-19 logical security state/outbound delivery.
Observability	bp_observability	Safe telemetry only; no authority decisions.

Appendix E - Worker / Broker Mapping Quick Reference

Question	Locked answer
Worker language	Python 3.13.x.
Execution library	Celery 5.6.x under strict DOC-18 adapter policy.
Source queues	Exactly six logical q18.* queues.
Job types	Exactly eleven canonical JOB18 types.
Control/dedupe	DOC-18 Redis 8 control plane via redis-py.
Broker	Separate Redis 8 plane via Celery/Kombu + queue adapter.
Far-future delay	Broker-private durable sorted scheduler storage; no reliance on Celery worker-memory ETA alone.
ACK	After side-effect commit or terminal stale/no-op only.
Retry	DOC-18 deterministic helper, not Celery default.
DLQ	Corresponding physical dlq18.* routing; replay revalidates tenant/state.
Schedule truth	PostgreSQL Post state/version/scheduled_for; 60-second site reconciliation remains mandatory.
Security state	Not stored in either Redis plane.

Appendix F - One-Page Developer Execution Reference

Area	Locked answer
REPO	One Git repo; pnpm-lock.yaml for JS/TS + uv.lock for Python; Turborepo top-level graph.
PUBLIC	14 current Next.js 16.3/React/TS SSR apps; site-local UI/shell/assets; no cross-site imports.
ADMIN	One React/Vite Admin; 19 modules/36 operational pages; auth surfaces not extra business pages.
API	One stateless Python/FastAPI app implementing exact DOC-15 RC7 OpenAPI; HTTP -> application -> ports/adapters.
WORKER	One Python Worker code app; Celery execution mapped to exact DOC-18 6 queues/11 jobs; custom broker delay adapter preserves far-future schedules.
DATABASE	PostgreSQL 17 + SQLAlchemy 2.0/Psycopg 3 access. Raw DOC-08 SQL migrations only. Site tx = BEGIN -> SET LOCAL app.site_id -> SQL -> COMMIT/ROLLBACK.
REDIS	Control/cache Redis for DOC-18 keys; separate durable broker Redis. Neither stores DOC-19 security state.
SEARCH	OpenSearch 3 derived index; opensearch-py; same-site; DOC-16 deterministic rank wins over provider defaults.
MEDIA	S3 private origin + boto3 + pyvips fixed derivatives; private original never public; no dynamic Next image transform.
SECURITY	argon2-cffi/PyOTP/opaque sessions/CSRF with DOC-19; separate conditional-TTL security store; Vault + SMTP.
SHARED	Allowed: contracts/generated clients/pure utils and backend Python core/adapters. Forbidden: shared public visual runtime/site mutable state/assets.
BUILD	One site builds independently. Python core fans out to API/Worker. OpenAPI/shared contract changes intentionally fan out.
STATUS	Architecture specified; implementation remains NOT_VERIFIED/EVIDENCE_REQUIRED.
DOWNSTREAM	DOC-24 RC1 must be revised to Next.js/Python/mixed-lock deployment implementation before global chain synchronization.